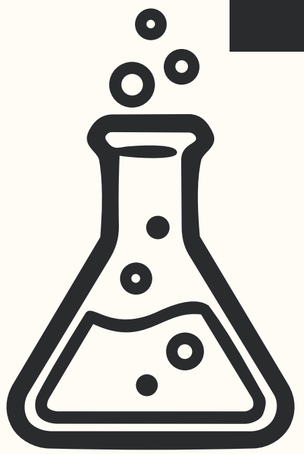
WORKSHOP #3:

HYPOTHESIS & METHODS



NEXTINRESEARCH 2026

Building a HYPOTHESIS

How to Hypothesis

- IF ___, THEN ____
- Evidence-backed ARGUMENT (good news, you should already had justification and background research!)
- Taking a position in your question (it doesn't have to be correct)

Easiest way to do this: take your focused research question, and rearrange it into this format!

GUIDED PRACTICE

EXAMPLE FROM LAST TIME: Is there a correlation between the weekly gaming time of Australian University students and their assessed physical activity levels as recorded through an International Physical Activity Questionnaire-Short Form?

If [Australian University students have a higher weekly gaming time], **then** [pysical activity levels as recorded through an International Physical Activity Questionnaire-Short Form will be lower, showing a direct correlation].

TURN AND TALK!

Research Question: How do AI-driven tutoring systems influence critical thinking skills in high school students?

Turn and talk! Transform the research question into a hypothesis!

TRY IT ON YOUR OWN

Take out your research question!

With your group or on your own, practice transform your research question into a hypothesis

Write at least TWO iterations on your worksheet, they can be the same or a different argument!

Outlining materials + methods

1. IDENTIFY "NEEDS"

Doing statistical test with existing data? → NEED a dataset

Creating a robotics/engineering prototype? → NEED specific building materials & quantity

Giving middle school students quiz/questionare as part of results? → need to MAKE the quiz

TURN AND TALK

What is a NEED for YOUR project? Write this down on your worksheet and discuss with your group or on your own!

2. "STEAL" FROM YOUR SOURCES

- Refer to your papers' MATERIALS AND METHODS, and see HOW they are finding data, tools used, HOW data was collected
- TAKE FROM MULTIPLE SOURCES, not just ONE (to avoid plagarism)
- TRACK THIS INFORMATION AND ORDER OF EXPERIMENTATION!

QUICK NOTE:

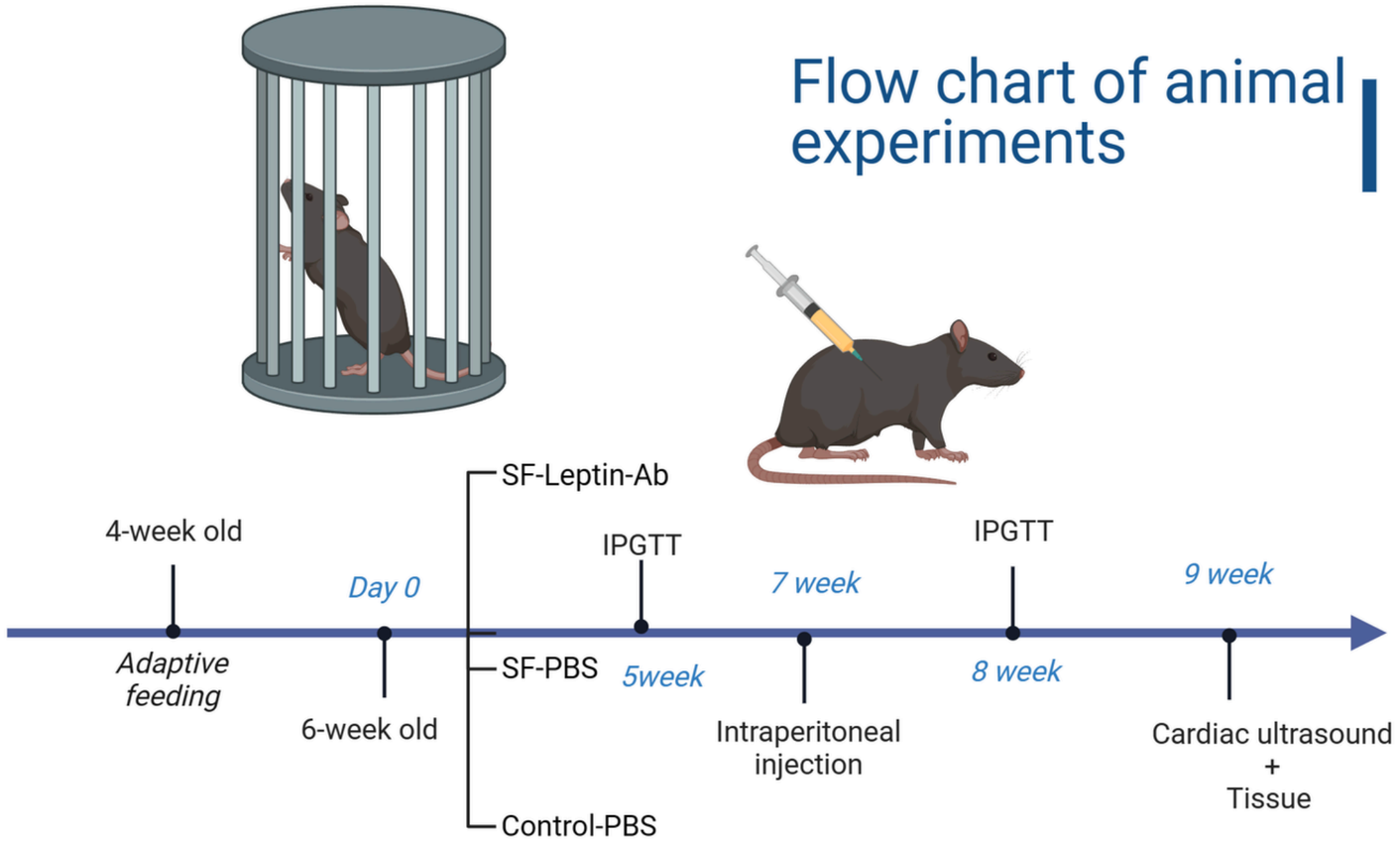
Remember:

- **Your “treatment” or “prototype” corresponds to your INDEPENDENT VARIABLE (what you are changing)**
- **Data COLLECTION is going to correspond with your DEPENDENT VARIABLE (what you’re measuring)**

3. IMPLEMENT YOUR "NOVEL" COMPONENT

- From your draft methodologies, add in your novel component!
- ex: a study uses Compound A to see if it WORKS, but now you want to test a CHEAPER Compound A or find a cost-efficient alternative

4. FLOW CHART(OPTIONAL)



- To visually confirm what you are doing
- Good for replicability/if someone ELSE tries to do your experiment

KEY COMPONENTS + TIPS:

Keep track of EVERYTHING:

- **Participants or Samples:** Describe how your subjects or samples were selected, recruited, or acquired. State sample sizes, demographics, and inclusion/exclusion criteria (who can/cannot be included in a study)
- **Materials and Equipment:** Give **EXACT** details to ensure your tools can be sourced. List manufacturer names, model numbers, catalog/lot numbers, and version numbers for any softwares used
- **Step-by-step Protocol:** Detail exactly how the experiment or study was conducted (temperatures, doses, survey formats, etc.)
- **Data and Statistical Analysis (NEXT LESSON!)**

FUN ACTIVITY!

ACTIVITY #1: ORIGAMI "MATERIALS AND METHODS"

How it will go:

- **We will count you off and split into two groups: one writer and one tester**
- **WRITERS: wil split and learn how to make a simple origami shape. Then, with this instruction you will have 8 minutes to write down a protocol for how to make it**
- **TESTERS: After the 8 minutes are done, you will test your WRITER's protocol, and try to recreate the shape using ONLY what is on the protocol your group member gives**

BEST ORIGAMI TO THE ORIGINAL WINS!

THANK YOU!